

Mushroom Edibility Classification: A Comparative Study of Tabular ML vs Image-Based CNN

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Abstract—Mushroom classification can have deadly consequences if done incorrectly when classifying for food consumption. The morphologies of edible and poisonous mushrooms overlap, and using simple decision rules is discouraged by field guides due to the risk of misclassification [7]. To support the binary task of classifying mushrooms as edible or poisonous, this paper compares six machine learning algorithms against a fine-tuned convolutional neural network (CNN), with the goal of comparing image-based classification against tabular classification restricted to only photographically visible features.

By restricting the tabular dataset to visually observable features, I attempt to give both modalities a fair comparison. The datasets used are the Secondary Mushroom Dataset [12] (61,069 samples, 173 species) and the DF20 Danish Fungi Dataset [10] (21,580 images, 45 overlapping species). Note that the two datasets are not combined, as there is no common feature to merge them on. Five features are removed from the tabular dataset prior to training due to missing values or lack of visual observability. Non-linear tabular classifiers (Random Forest, k-NN, MLP) achieve near-perfect accuracy of 99.99%, while ResNet18 reaches 83.37%. This gap may be attributed to visual ambiguity between morphologically similar look-alike species, or limitations in the training dataset or CNN. These distinctions may be captured by symbolic feature descriptions that photographs cannot convey. The CNN also produces 254 false negatives, misclassifying poisonous mushrooms as edible, the worst-case outcome for an edibility classifier.

Index Terms—Mushroom classification, edibility prediction, convolutional neural networks, random forests, ResNet, one-hot encoding, binary classification, support vector machines, ensemble methods, deep learning.

I. INTRODUCTION

Recently, mushroom foraging has been supplemented by smartphone applications that claim to identify mushroom edibility from photographs in accuracy ranges of 89–100% [5]. Despite their wide adoption and claimed accuracy, such tools carry substantial safety risks. Li et al. [6] report that mushroom poisoning caused thousands of incidents and hundreds of deaths in China alone between 2010 and 2020, the vast majority from misidentification of deadly species such as *Amanita phalloides* (Death Cap) and *Galerina marginata*. The UCI Primary Mushroom Dataset [11] notes explicitly that “there is no simple rule for determining the edibility of a mushroom”.

The primary limitation of the 1987 UCI Primary Mushroom dataset is its scope: 8,124 hypothetical samples from only 23 species of two genera, with near-linear separability. This small sample size and limited species inclusion fails to accurately represent the real-world challenges of mushroom classification. The Secondary Mushroom Dataset [12], developed in 2021, was introduced specifically to address these

limitations, covering 173 species across 23 families with non-linear separability and significant missing data. This tabular dataset was created synthetically to mimic the measurements that would be taken by mycologists when classifying real-world collected mushrooms in a lab-based setting. Although synthetic, the dataset only includes characteristic values that represent real-world mushroom species.

Image-based mushroom identification has recently been studied through deep learning techniques. Picek et al. [10] demonstrate competitive species identification on the DF20 Danish Fungi Dataset using CNN architectures, though their evaluation targets multi-class species recognition rather than binary edibility prediction. They also note that the DF20 dataset has no overlap with ImageNet, an image database used to train ResNet CNN models. More recently, Mir et al. [8] propose a combined CNN and Random Forest framework for mushroom disease diagnosis, further demonstrating the applicability of hybrid approaches to fungal classification. Paudel and Bhatta [9] show that Random Forest outperforms alternative classifiers on the UCI dataset, consistent with broader findings that tree-based models perform better on tabular data [3].

This paper poses the following research question: *how does CNN image classification compare to tabular ML classification when both are restricted to the same visually observable features?* To answer this fairly, each of the 20 features in the Secondary dataset is classified by visual observability and those that cannot be inferred from a photograph are removed. Six tabular classifiers spanning four machine learning methodologies are then trained, and a ResNet18 CNN is fine-tuned on a species-matched image dataset. Their performance on classifying mushrooms is compared and the safety implications of the results are discussed.

II. BACKGROUND

A. UCI Mushroom Classification Datasets

The UCI Primary Mushroom Dataset [11] contains 8,124 samples from 23 species of the *Agaricus* and *Lepiota* genera. The dataset is near-linearly separable with the *odor* feature alone achieving 98.5% classification accuracy. Because of this, this dataset is not representative of real-world difficulty and was rejected for use in this study.

Wagner et al. [12] introduced the Secondary Mushroom Dataset to address the Primary’s limitations. Simulated from species-level attribute distributions across 173 species and 23 families, it is non-linearly separable, contains significant missing data, and creates a more complex classification challenge. The dataset is described as a more realistic benchmark

for evaluating the limits of machine learning approaches to mushroom edibility classification. In this project, Logistic Regression achieves only 76.8% accuracy on this dataset, showing that the dataset does not have a linearly separable structure.

B. Tabular vs Image Classification

Grinsztajn et al. [3] provide an analysis demonstrating that tree-based models tend to outperform deep learning methods on medium-sized tabular datasets. They attribute this to the irregular target function structure typical of tabular tasks, which tree-based axis-aligned splits consistently navigated more effectively than continuous deep learning decision surfaces did for their tests.

Transfer learning with CNNs has shown strong performance across fine-grained visual recognition benchmarks [4]. Therefore, ResNet18 pretrained on the 1.2 million images in ImageNet was chosen for classifying the DF20 image dataset.

C. DF20 Danish Fungi Dataset

The Danish Fungi 2020 (DF20) dataset [10] is a large-scale citizen science collection of 295,938 mushroom photographs across 1,604 fungal species, gathered by amateur naturalists through the Danish Fungal Atlas platform. It was constructed as a benchmark for the FungiCLEF fine-grained species recognition competition series, targeting multi-class species identification rather than binary edibility prediction. Images reflect real-world field photography conditions, varying widely in quality, angle, and background. This project repurposes DF20 for binary edibility classification, filtering it to species that overlap with the Secondary dataset and assigning edibility labels from mycological literature.

III. DATASET AND EXPLORATORY ANALYSIS

A. Secondary Mushroom Dataset

The Secondary Mushroom Dataset [12] is sourced from the UCI Machine Learning Repository (id=848). It contains 61,069 samples described by 17 categorical and 3 continuous features with a binary edibility target. The dataset covers 173 mushroom species from 23 families, synthetically generated from species descriptions in the book *Mushrooms and Toadstools* for species commonly found in Europe. The class balance is 27,181 edible (44.5%) and 33,888 poisonous (55.5%), shown in Fig. 1.

Nine features have missing values, ranging from 4.0% (ring-type) to 94.8% (veil-type), shown in Fig. 2. Features exceeding a 50% missingness threshold are dropped: veil-type (94.8%), spore-print-color (89.6%), veil-color (87.9%), stem-root (84.4%), and stem-surface (62.4%). The last two are also unobservable and would have been removed regardless.

B. Feature Observability Classification

A key component of this comparison is the binary classification of each feature by visual observability: *observable* (visible or inferable from a photograph) or *not observable* (requires physical handling, laboratory testing, or cannot be

inferred from an image). Table II presents this classification for all 20 original features.

The five not-observable features (does-bruise-or-bleed, stem-root, spore-print-color, habitat, season) are removed prior to training. The rationale is that a CNN operating on a mushroom photograph has no access to these features, so including them in the tabular model would grant it an unfair informational advantage. After removing unobservable and mostly missing features, and applying one-hot encoding, the tabular feature space comprises 3 continuous and 9 categorical features expanding to 78 dimensions. Notably, all 173 species in the tabular dataset are species that contain stems and caps in their fruiting bodies. All of the features remaining after processing would be relatively distinguishable simply by looking at the mushroom.

C. DF20 Image Dataset

DF20 is filtered to the 45 species present in both DF20 and the Secondary dataset's 173-species taxonomy, subject to a minimum of 50 images per species. The 50-image threshold ensures each species contributes a statistically meaningful number of samples to the test partition. This reduces the dataset from 295,938 images across 1,604 species to 21,580 images across 45 species: 22 edible and 23 poisonous, for a near-balanced class distribution of 48.9%/51.1%.

The dataset is stratified split per species into 70% training (15,146), 15% validation (3,217), and 15% test (3,217). Per-species stratification differs from the class-label stratification used in the tabular track and is necessary here because species vary widely in image count (56–1,472): splitting by class label alone could leave rare species entirely absent from the test set.

Edibility labels are assigned through a manually curated mapping (edibility_map.py) derived from MycoBank, GBIF toxicity records, and field guides [7]. All non-edible categories (toxic, mildly poisonous, and conditionally edible) are mapped to poisonous (label 1), to maintain a safety-first methodology. The complete species-to-label mapping is provided in Table I.

D. Exploratory Data Analysis

Fig. 1 shows the class distribution of the Secondary dataset (44.5% edible, 55.5% poisonous), motivating stratified splits throughout. Fig. 2 shows the nine features with missing data, with five columns exceeding 50% missingness. Fig. 3 shows continuous feature distributions by class with substantial overlap between edible and poisonous distributions across all three size features, indicating low individual discriminative power for measurements alone. Fig. 4 shows class-conditional distributions for selected categorical features, where cap-color and gill-color exhibit strong class-conditional signals, motivating their retention as primary visual features.

IV. MODELS AND METHODOLOGY

A. Tabular Preprocessing

Raw features are loaded via ucimlrepo and processed through a ColumnTransformer pipeline. Continuous features (cap-diameter, stem-height, stem-width) are standardized using

TABLE I
EDIBILITY LABELS FOR THE 45 DF20 SPECIES OVERLAPPING WITH THE SECONDARY MUSHROOM DATASET. ALL NON-EDIBLE CATEGORIES ARE MAPPED TO POISONOUS (LABEL 1).

Species	Class	Label	Species	Class	Label
<i>Agaricus augustus</i>	Edible	0	<i>Laccaria amethystina</i>	Edible	0
<i>Agaricus campestris</i>	Edible	0	<i>Laccaria laccata</i>	Edible	0
<i>Agaricus xanthodermus</i>	Poisonous	1	<i>Lactarius deliciosus</i>	Edible	0
<i>Amanita citrina</i>	Poisonous	1	<i>Lactarius torminosus</i>	Poisonous	1
<i>Amanita muscaria</i>	Poisonous	1	<i>Lepiota cristata</i>	Poisonous	1
<i>Amanita pantherina</i>	Poisonous	1	<i>Lepista nuda</i>	Edible	0
<i>Amanita phalloides</i>	Poisonous	1	<i>Macrolepiota procera</i>	Edible	0
<i>Amanita rubescens</i>	Poisonous	1	<i>Marasmius oreades</i>	Edible	0
<i>Amanita virosa</i>	Poisonous	1	<i>Morchella esculenta</i>	Edible	0
<i>Armillaria mellea</i>	Poisonous	1	<i>Mycena galericulata</i>	Poisonous	1
<i>Boletus edulis</i>	Edible	0	<i>Paxillus involutus</i>	Poisonous	1
<i>Cantharellus cibarius</i>	Edible	0	<i>Pleurotus ostreatus</i>	Edible	0
<i>Clitocybe nebularis</i>	Poisonous	1	<i>Russula cyanoxantha</i>	Edible	0
<i>Clitocybe odora</i>	Edible	0	<i>Russula emetica</i>	Poisonous	1
<i>Coprinus comatus</i>	Edible	0	<i>Russula ochroleuca</i>	Poisonous	1
<i>Cortinarius rubellus</i>	Poisonous	1	<i>Suillus luteus</i>	Edible	0
<i>Craterellus cornucopioides</i>	Edible	0	<i>Tricholoma equestre</i>	Poisonous	1
<i>Flammulina velutipes</i>	Edible	0	<i>Tricholoma portentosum</i>	Edible	0
<i>Galerina marginata</i>	Poisonous	1	<i>Tricholoma terreum</i>	Edible	0
<i>Grifola frondosa</i>	Edible	0			
<i>Gyromitra esculenta</i>	Poisonous	1			
<i>Hebeloma crustuliniforme</i>	Poisonous	1			
<i>Hydnum repandum</i>	Edible	0			
<i>Hygrocybe conica</i>	Poisonous	1			
<i>Hypholoma fasciculare</i>	Poisonous	1			
<i>Inocybe geophylla</i>	Poisonous	1			

22 edible · 23 poisonous · 45 total

TABLE II
FEATURE OBSERVABILITY CLASSIFICATION. FEATURES MARKED NO ARE REMOVED BEFORE TRAINING.

Feature	Type	Obs.	Note
cap-diameter	Continuous	Yes	Needs scale ref.
stem-height	Continuous	Yes	Needs scale ref.
stem-width	Continuous	Yes	Needs scale ref.
cap-shape	Categorical	Yes	Visible from above
cap-surface	Categorical	Yes	Texture in photo
cap-color	Categorical	Yes	Primary visual cue
gill-attachment	Categorical	Yes	Underside photo
gill-spacing	Categorical	Yes	Underside photo
gill-color	Categorical	Yes	Underside photo
stem-color	Categorical	Yes	Fully visible
has-ring	Categorical	Yes	Clearly visible
ring-type	Categorical	Yes	Visible if present
does-bruise-or-bleed	Categorical	No	Requires handling
stem-root	Categorical	No	Underground
spore-print-color	Categorical	No	Requires lab test
habitat	Categorical	No	Not reliably captured
season	Categorical	No	Cannot infer from image

Dropped in preprocessing (not in training):

veil-type	Categorical	Yes	94.8% missing
veil-color	Categorical	Yes	87.9% missing
stem-surface	Categorical	Yes	62.4% missing

StandardScaler (zero mean, unit variance). Categorical features are one-hot encoded using OneHotEncoder with handle_unknown='ignore' to manage unseen values at inference. Encoded values are clipped to $[-10, 10]$ to prevent numerical overflow during Logistic Regression optimization on the high-dimensional encoded feature space. Missing categorical values are imputed with the column mode, and continuous missing values are imputed with the column median.

Class Distribution — Secondary Mushroom Dataset

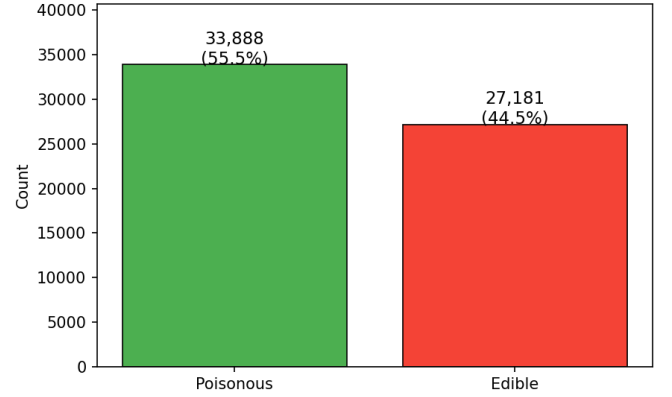


Fig. 1. Class distribution of the Secondary Mushroom Dataset: 27,181 edible (44.5%) and 33,888 poisonous (55.5%).

The dataset is split into 70% training (42,747), 15% validation (9,161), and 15% test (9,161) using stratified sampling on the class label to preserve the 55.5%/44.5% class ratio in each partition. The validation set is used exclusively for MLP early stopping and learning rate scheduling and is never used for weight updates.

B. Tabular Models

Six classifiers spanning four tiers are evaluated.

1) *Logistic Regression (Baseline)*: The saga solver is selected for sparse, high-dimensional data. Regularization is strengthened ($C=0.1$ vs. default $C=1.0$) to prevent weight inflation on the 78-dimensional encoded features. A maximum

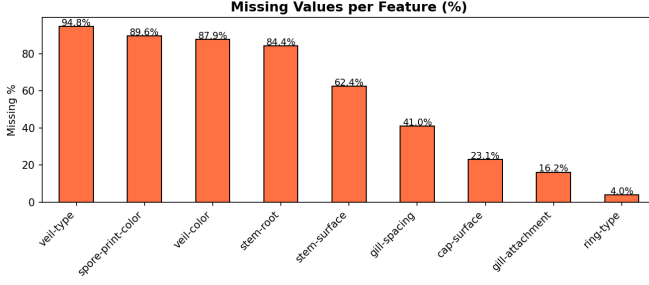


Fig. 2. Missing value percentages per feature. Features exceeding 50% (veil-type, spore-print-color, veil-color, stem-root, stem-surface) are dropped prior to training.

of 5,000 iterations ensures convergence on the 42,747-sample training set. This model serves as a linear baseline, and its limited performance confirms non-linear separability.

2) *Support Vector Machine*: An SVM with an RBF kernel maps the feature space to an infinite-dimensional reproducing kernel space, enabling non-linear class boundaries without explicit feature expansion. `probability=True` enables Platt scaling for ROC-AUC computation.

3) *k-Nearest Neighbours*: k -NN with $k = 5$ uses Euclidean distance in the 78-dimensional encoded space. `StandardScaler` is applied within a Pipeline to prevent large-scale continuous features from dominating distance computations.

4) *Random Forest*: A Random Forest of 200 fully-grown decision trees is trained, with bagging and random feature subsampling providing regularization [1]. No maximum depth is set, allowing trees to grow until leaves are pure. Feature importances are recorded from Gini impurity for post-hoc analysis.

5) *XGBoost*: XGBoost [2] is trained with 200 gradient-boostered trees, learning rate 0.1, maximum depth 4, and L2 regularization ($\lambda = 1$, default). The logloss evaluation metric suppresses verbose output during training.

6) *Multilayer Perceptron (PyTorch)*: A feedforward MLP with architecture $78 \rightarrow 256 \rightarrow 128 \rightarrow 64 \rightarrow 1$ is implemented in PyTorch. `BatchNorm1d` and `Dropout(0.3)` follow each hidden layer. The network is trained for 60 epochs using Adam ($\text{lr}=10^{-3}$, $\text{weight_decay}=10^{-4}$) and binary cross-entropy loss. A `ReduceLROnPlateau` scheduler halves the learning rate after 5 epochs of non-improving validation loss.

C. CNN Pipeline

ResNet18 [4] pretrained on ImageNet-1k is fine-tuned for binary edibility classification. The final fully-connected layer (`Linear(512→1000)`) is replaced with a binary classification head: `Dropout(0.3) → Linear(512→1) → Sigmoid`. All 11.2 million parameters are updated during training (full fine-tuning, not frozen backbone).

Training images undergo augmentation: resize to 256×256 , random crop to 224×224 , random horizontal and vertical flips, color jitter (brightness ± 0.3 , contrast ± 0.3 , saturation ± 0.2 , hue ± 0.05), and random rotation ($\pm 15^\circ$). Validation and test images are deterministically resized to 224×224 . All images

TABLE III
5-FOLD STRATIFIED CROSS-VALIDATION RESULTS (TRAINING SET ONLY).
MLP EVALUATED ON TEST SET ONLY.

Model	Acc.	$\pm$ Std	F1	AUC
Random Forest	0.9999	0.0001	0.9999	1.0000
k-NN ($k=5$)	0.9997	0.0001	0.9997	1.0000
SVM (RBF)	0.9922	0.0015	0.9930	0.9986
XGBoost	0.9814	0.0011	0.9832	0.9982
MLP (PyTorch)	—	—	—	—
Logistic Regression	0.7679	0.0057	0.7890	0.8364

are normalized using ImageNet statistics (mean=[0.485, 0.456, 0.406], std=[0.229, 0.224, 0.225]).

The model is trained for 20 epochs using Adam ($\text{lr}=10^{-4}$, $\text{weight_decay}=10^{-4}$) and binary cross-entropy loss, with `ReduceLROnPlateau` (patience=3, factor=0.5). `WeightedRandomSampler` is applied to the training `DataLoader` to ensure balanced class sampling per batch. Training completes in approximately 56 minutes (batch size 32).

D. Evaluation Protocol

Tabular models are evaluated using 5-fold stratified cross-validation on the training set, followed by final evaluation on the held-out test set of 9,161 samples. The CNN is evaluated on a separate held-out test set of 3,217 DF20 images. Both test sets are withheld from all training and hyperparameter selection. Reported metrics are accuracy, precision, recall, F1-score (weighted), and ROC-AUC. The primary safety metric is recall on the poisonous class: a false negative would predict a poisonous mushroom as edible, which constitutes a potentially fatal error.

V. RESULTS

A. Cross-Validation Results

Table II presents 5-fold stratified cross-validation results for all tabular models on the training set. Ensemble models achieve near-perfect mean accuracy with low standard deviation across folds, showing stable generalization. Logistic Regression’s 76.8% cross-validation accuracy confirms non-linear separability, consistent with Wagner et al.’s design intent [12]. The MLP was evaluated on the test set only due to training time constraints.

B. Test Set Results

Table III presents the primary results comparing all tabular models and ResNet18 on their respective held-out test sets. Random Forest and MLP achieve near-perfect accuracy of 99.99%, with MLP attaining a perfect AUC of 1.000. k-NN achieves 99.97% with perfect AUC. SVM achieves 99.53% with perfect recall on the poisonous class, meaning it produces zero false negatives at a cost of slightly reduced precision (0.9916). XGBoost achieves 98.97%. Logistic Regression achieves 76.79%, confirming that the observable feature set is non-linearly separable.

ResNet18 achieves 83.37% accuracy with F1 of 0.8408, AUC of 0.9188, precision of 0.8341, and recall of 0.8476

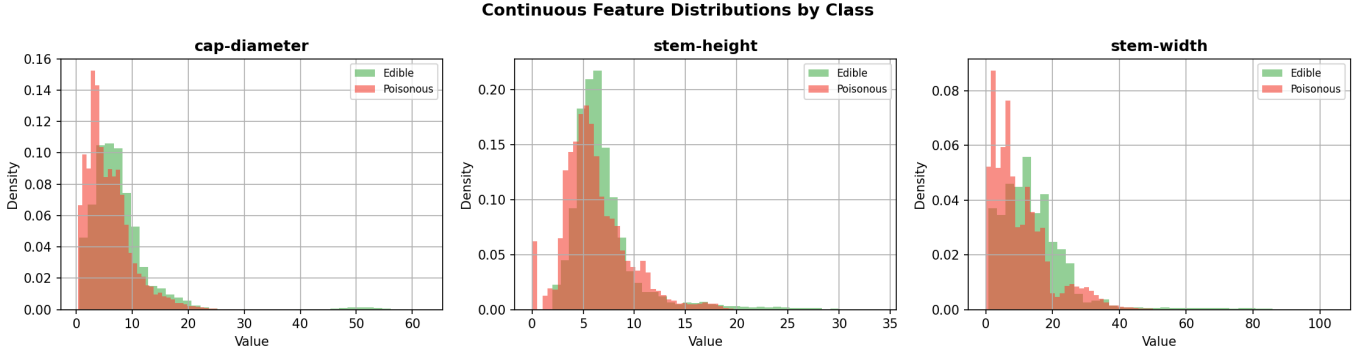


Fig. 3. Normalized distributions of cap-diameter, stem-height, and stem-width by class. Substantial overlap indicates low individual discriminative power for size features alone.

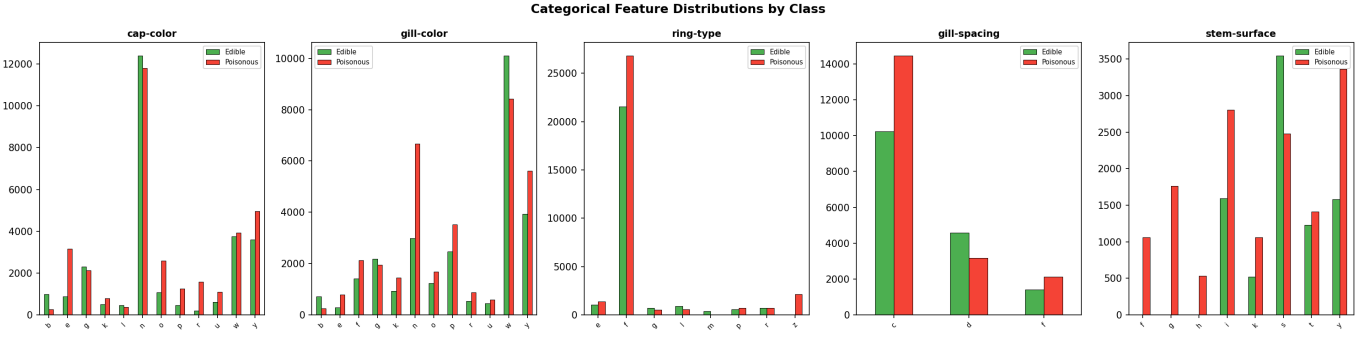


Fig. 4. Class-conditional distributions for selected categorical features. Cap-color and gill-color exhibit strong class-conditional signals.

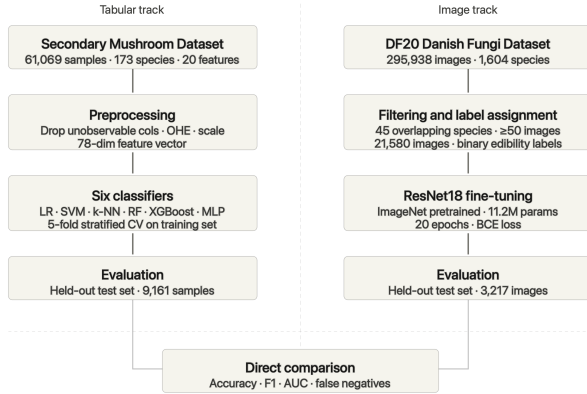


Fig. 5. Project pipeline. Both tracks share a held-out evaluation stage and converge at the direct comparison step.

on the DF20 test set. The accuracy gap between the best tabular model and the CNN is 16.62 percentage points. The full comparison is shown in Fig. 6. SVM achieves perfect recall on the poisonous class while the CNN misses 15.24% of poisonous samples.

C. Consequence Analysis

The CNN produces 254 false negatives on the test set: poisonous mushrooms incorrectly classified as edible. This represents 15.2% of the 1,667 poisonous test samples. False

TABLE IV
TEST SET RESULTS. CNN TEST SET IS SEPARATE (3,217 DF20 IMAGES VS. 9,161 TABULAR SAMPLES).

Model	Type	Acc.	F1	AUC	Prec.	Rec.
Random Forest	Ensemble	0.9999	0.9999	1.0000	1.000	0.9998
MLP (Neural Net)	Deep Learn	0.9999	0.9999	1.000	1.000	0.9998
k-NN ($k=5$)	Classical	0.9997	0.9997	1.000	1.000	0.9994
SVM (RBF)	Classical	0.9953	0.9958	0.9995	0.9916	1.000
XGBoost	Ensemble	0.9897	0.9907	0.9991	0.9917	0.9898
Logistic Reg.	Baseline	0.7679	0.7900	0.8361	0.7936	0.7864
ResNet18 (CNN)	Image	0.8337	0.8408	0.9188	0.8341	0.8476

positives (edible classified as poisonous) number 281. By contrast, the best tabular models produce near-zero false negatives: Random Forest produces 1, MLP produces 1, and SVM produces 0. These results are shown in Figs. 8 and 9.

D. CNN Training Analysis

ResNet18 validation accuracy improves from 71.1% at epoch 1 to a peak of 84.2% at epoch 18. The learning rate scheduler triggers at epoch 16, reducing the rate from 10^{-4} to 5×10^{-5} after three epochs of non-improving validation loss, producing a visible acceleration in both curves. Training loss decreases from 0.557 to 0.133 over 20 epochs. The best checkpoint, saved at epoch 18 (minimum validation loss 0.391), is used for all reported results. The training curves are shown in Fig. 10.

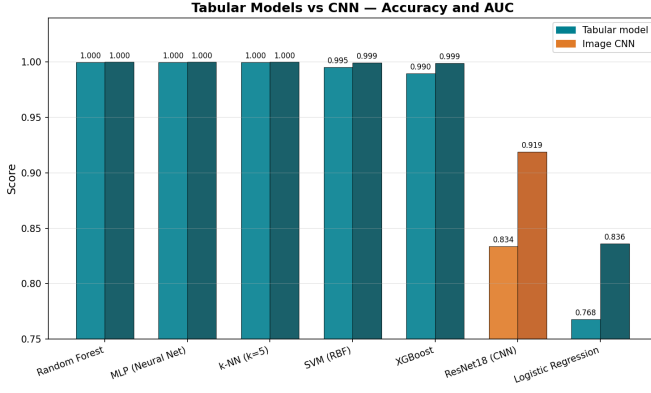


Fig. 6. Accuracy (lighter) and ROC-AUC (darker) for all models. Tabular models (teal) achieve near-perfect scores; ResNet18 CNN (orange) shows a 16.6 percentage point accuracy gap.

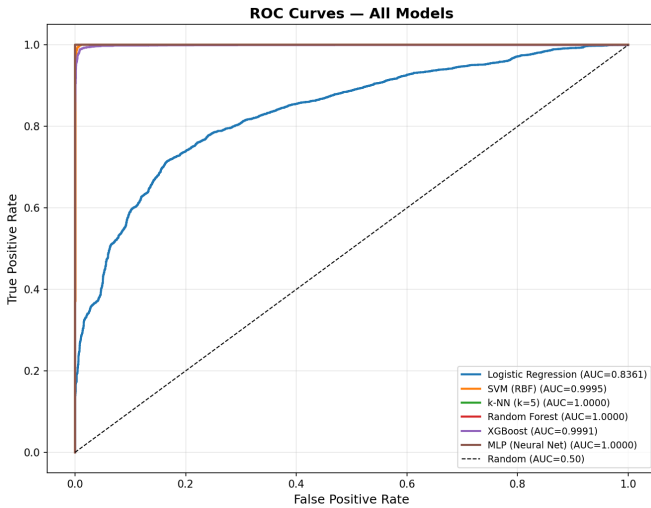


Fig. 7. ROC curves for all tabular models. Random Forest, MLP, and k-NN achieve AUC of 1.000. Logistic Regression AUC of 0.836 shows non-linear separability.

E. Feature Importance Analysis

Random Forest Gini importance and XGBoost gain importance both identify cap-surface as the most important individual OHE feature, followed by stem-color, cap-color, and gill-color. These features are grouped by original feature name for interpretability. Notably, all four highest-ranked features are fully visually observable, which partially explains why the CNN achieves substantial accuracy despite the 16.6% gap. The three continuous size features (cap-diameter, stem-height, stem-width) have moderate importance but are only partially observable without a scale reference. Feature importances are visualized in Fig. ??.

VI. DISCUSSION

A. Tabular Models Achieve Near-Perfect Accuracy

The secondary dataset, while substantially harder than the Primary UCI dataset, is learnable by non-linear classifiers with sufficient training data. Random Forest, k-NN, and MLP all achieve 99.99% accuracy, confirming that the observable

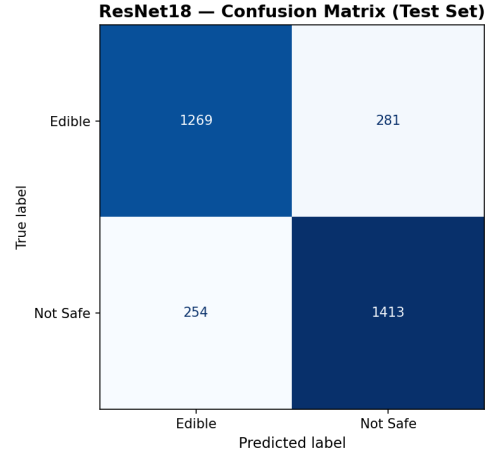


Fig. 8. ResNet18 confusion matrix. 254 false negatives (poisonous predicted edible, 15.2% of poisonous samples). False positives: 281.

feature set carries sufficient discriminative information for near-perfect edibility classification. Tree-based models are particularly well-suited to this task: the irregular, high-interaction decision boundaries that characterize mycological classification are well-captured by axis-aligned splits, consistent with the findings of Grinsztajn et al. [3].

The failure of Logistic Regression (76.8%) demonstrates that the observable feature decision boundary is non-linear and cannot be adequately estimated by a hyperplane in the 78-dimensional encoded feature space. This validates the choice of the Secondary dataset as a more demanding benchmark.

The SVM result shows perfect recall on the poisonous class with zero false negatives. The RBF kernel’s capacity for non-linear boundaries in high-dimensional space allows it to identify a decision surface that correctly flags every poisonous sample in the test set, at the cost of slightly elevated false positives (43 edible samples classified as poisonous). For someone trying to maintain safety above all else, this tradeoff is strongly preferable.

B. CNN At Only 83.4%

Both modalities are restricted to the same visually observable features, so the accuracy gap might not be attributed to information asymmetry. However, it is difficult to say that the features in each dataset are truly equal, as the different datasets may have unseen confounding variables affecting this comparison. One explanation is visual ambiguity, where morphologically similar look-alike species share identical or near-identical visual appearances but differ in toxicity. In the tabular dataset, each feature has high precision with very little noise. In the image dataset, features are not clearly delineated and the classifier must infer cap-color, gill-color, and ring-type from pixel distributions that vary with lighting, angle, and photographic quality. Another explanation is that the tabular dataset contained 173 species as compared to DF20’s 45 represented species. The 45 species used in the DF20 may have been especially difficult to classify, and in the tabular dataset the 173 species spread may have hidden those difficulties by providing a more species-diverse training set.

Confusion Matrices — Test Set

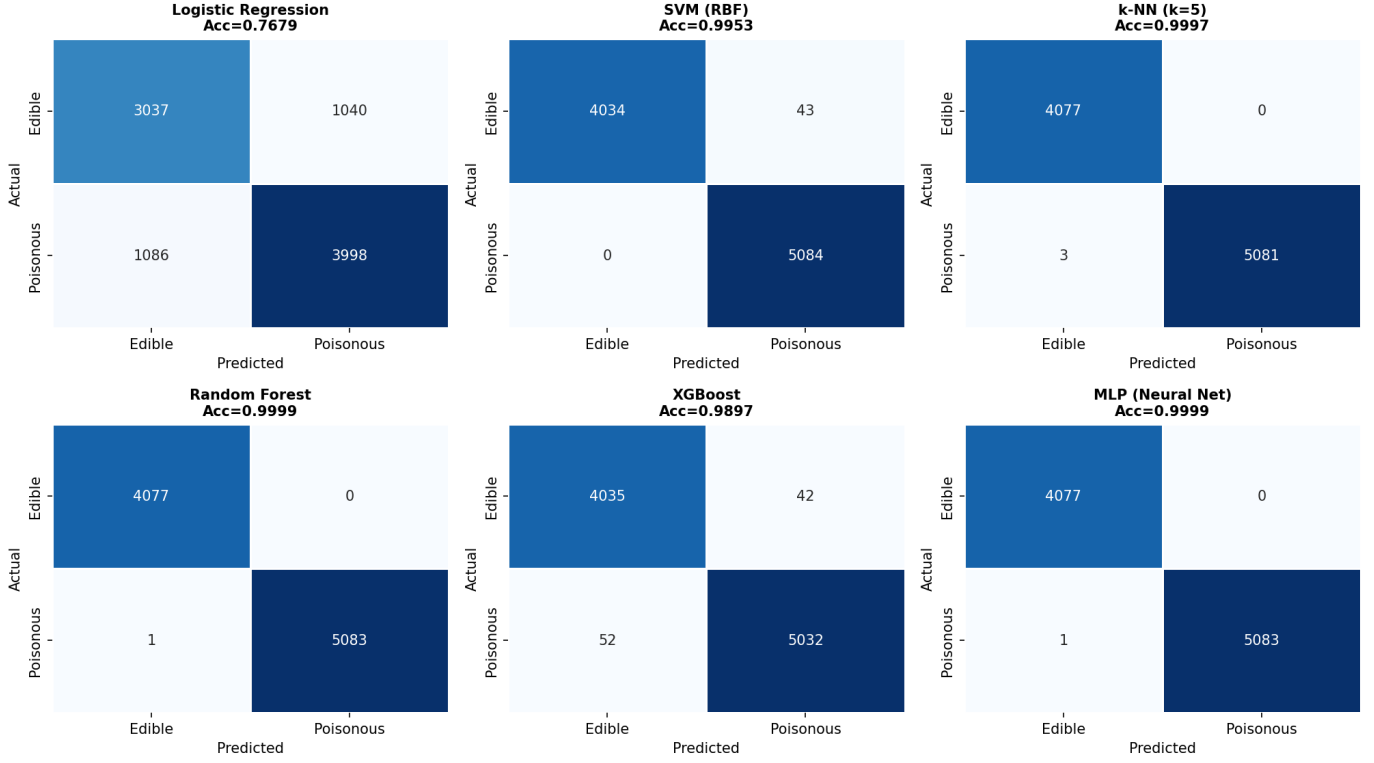


Fig. 9. Confusion matrices for all tabular models. Non-linear classifiers (Random Forest, MLP, k-NN) produce near-zero false negatives. Logistic Regression shows substantially higher error rates on both classes.

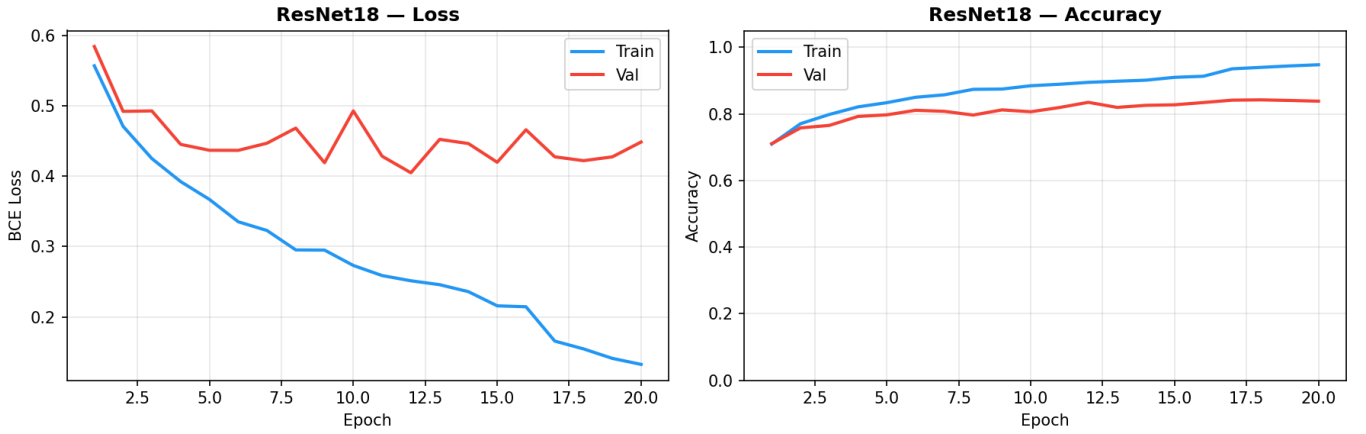


Fig. 10. ResNet18 training and validation loss (left) and accuracy (right) over 20 epochs. LR scheduler triggers at epoch 16. Best checkpoint saved at epoch 18.

The CNN ROC-AUC of 0.9188 is notably high despite the lower accuracy, indicating that the model has learned some useful visual features and can discriminate between classes across a range of thresholds. The issue is not that the CNN fails to learn, but that certain species pairs may be visually inseparable at the image level regardless of model architecture or training data size. This may place a ceiling on achievable CNN accuracy that the tabular model does not face, but further hyperparameter tuning tests would need to be done to examine

this.

C. Safety Implications

The 254 false negatives produced by the CNN represent a high safety concern: at the 83.4% accuracy reported here, approximately one in seven poisonous mushrooms would be misclassified as edible by this model. By contrast, the best tabular models produce between 0 and 1 false negatives on the same size test partition. This asymmetry may show that

symbolic tabular feature descriptions encode discriminative information that photographs cannot convey, and that this particular image-based mushroom identification application relying on CNN classification alone should not be trusted for edibility decisions.

D. Limitations

- The DF20 dataset is primarily Danish citizen science; the tabular dataset is synthetically generated from European species descriptions. Geographic distribution mismatch may introduce bias in both training distributions.
- The 45-species overlap represents only 26% of the 173 species in the Secondary dataset. Tabular models are trained on all 173 species; the CNN cannot be evaluated on the remaining 128.
- Species labels for DF20 are assigned by mycologists, but human error may result in some images being mislabelled.
- Ideally, a dataset would exist with both image and tabular data per sample, enabling a fully matched comparison. No such dataset currently exists at this scale.

VII. CONCLUSION

This paper presents a comparative study of tabular machine learning and CNN image classification for mushroom edibility prediction, with both modalities restricted to visually observable features. Three findings emerge.

First, non-linear tabular classifiers (Random Forest, MLP, k-NN) achieve near-perfect accuracy of 99.99% on observable features, confirming that the Secondary dataset is learnable and that ensemble methods are well-suited to its non-linear structure. Logistic Regression’s 76.8% accuracy confirms genuine non-linear separability, distinguishing this benchmark from the trivially-solvable Primary UCI dataset. SVM achieves perfect recall on the poisonous class with zero false negatives.

Second, ResNet18 achieves 83.37% accuracy, yielding a 16.6 percentage point gap. This underperforms as compared to ML methods on classifying from tabular data, indicating that evaluation on tabular data for these datasets is more reliable.

Third, the CNN produces 254 false negatives (15.2% of poisonous test samples), misclassifying potentially deadly mushrooms as edible. This safety finding reaffirms concerns for the deployment of image-based mushroom identification applications, which should be validated as safe before deployment.

Future work should investigate multi-modal approaches combining image features with contextual metadata (location, substrate, season), explore generalization to non-European species, and establish datasets with both tabular and image features for a proper comparison.

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